

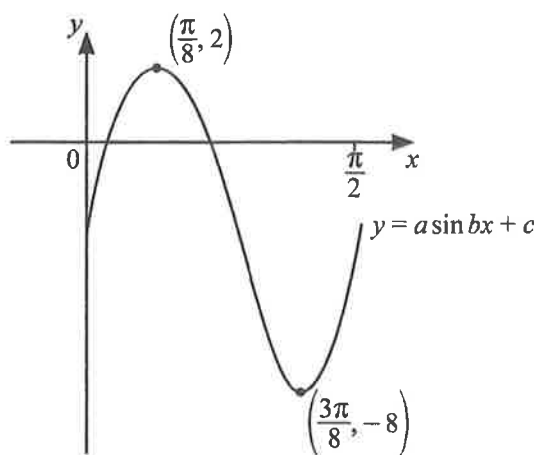
Basic

1. Given that θ is acute and $\cos \theta = c$, express, in terms of c ,
- (i) $\tan \theta$, [3]
 - (ii) $\operatorname{cosec} \theta$. [1]
- [N19/I/1]

Intermediate

2. (a) Find the amplitude and period of
- (i) $2 \sin x$, [1]
 - (ii) $3 - 4 \cos 2x$. [2]
- (b) Sketch, on the same diagram, the curves $y = 2 \sin x$ and $y = 3 - 4 \cos 2x$ for $0 \leq x \leq 2\pi$ radians. [3]
- [N22/I/10]

3.



The diagram shows the curve $y = a \sin bx + c$ for $0 \leq x \leq \frac{\pi}{2}$ radians. The curve has a maximum point at $(\frac{\pi}{8}, 2)$ and a minimum point at $(\frac{3\pi}{8}, -8)$.

- (a) Explain why $c = -3$. [2]
 - (b) Explain why $b = 4$. [2]
 - (c) Hence find the equation of the curve. [2]
- [N21/I/7]
4. It is given that $f(x) = 2 \sin 2x$ and $g(x) = 3 \cos \left(\frac{x}{2}\right) - 1$.
- (i) State the least and greatest values of $f(x)$. [1]
 - (ii) State the least and greatest values of $g(x)$. [2]
 - (iii) State the period of $f(x)$. [1]
 - (iv) State the period of $g(x)$. [1]
 - (v) Sketch, on the same axes, the graphs of $y = f(x)$ and $y = g(x)$ for $0^\circ \leq x \leq 360^\circ$. [4]
 - (vi) State the number of solutions of the equation $2 \sin 2x + 1 = 3 \cos \left(\frac{x}{2}\right)$ for $0^\circ \leq x \leq 360^\circ$. [1]
- [N19/I/12]

5. (a) State the values between which each of the following must lie:

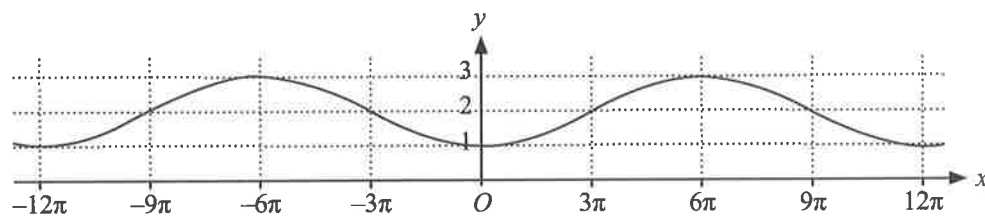
(i) the principal value of $\sin^{-1} x$,

[1]

(ii) the principal value of $\cos^{-1} x$.

[1]

(b)



The figure shows part of the graph of $y = a \cos \frac{x}{b} + c$. Find the value of each of the constants a , b and c . [3]

[N16/I/3]